Graph Theory Notes

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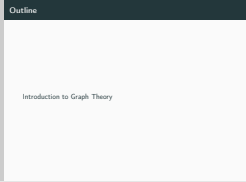
Outline

Introduction to Graph Theory



Introduction to Graph Theory

Networks and Relationships

- Much like group theory is the mathematical language of symmetry, graph theory provides languages for expressing relationships

Applications of graph theory

- Modeling networks
- Chemistry
- Machine learning
- Routing problems
- CYOA

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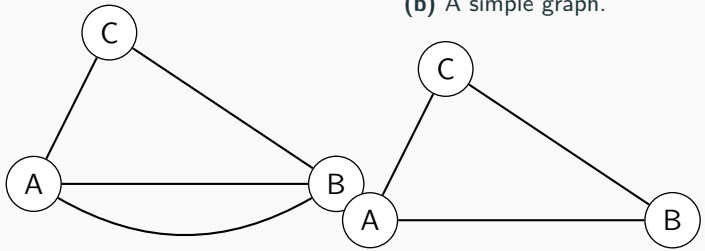
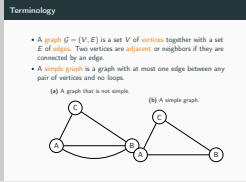
Terminology

- A graph $\mathcal{G} = (V, E)$ is a set V of vertices together with a set E of edges. Two vertices are adjacent or neighbors if they are connected by an edge.
- A simple graph is a graph with at most one edge between any pair of vertices and no loops.

(a) A graph that is not simple.

(b) A simple graph.





Friendship

- A friendship graph is a graph where each pair of vertices have exactly one neighbor (i.e., the "friend") in common.

Example (Friendship graphs)

Draw examples of friendship graphs.

Degrees, Paths and Cycles

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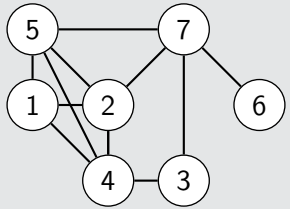
An Example

Example (Finding degrees, paths and cycles)

Find the following using the graph below:

1. The degree of each vertex.
2. The number of paths from vertex 1 to vertex 6.
3. The length of the shortest path from vertex 1 to vertex 6.

Figure 2: A graph that is not simple.



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An Application

Example (Scheduling tournaments)

Suppose six teams are playing in a tournament and each team must play exactly three others. How can this be done? What if only five teams are playing?

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An Impossibility Proof

Example (Not scheduling tournaments)

Prove that it's impossible to schedule a tournament for five teams such that each team plays exactly three others.

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Look at previous examples and try to find a relationship between degrees and edges.

Another Matching Problem

Example (Utility graph)

Three houses A, B and C must be connected to three utilities E (electric), G (gas) and W (water). Each house must connected to each utility exactly once. What graph represents this situation?

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Bipartite Graphs

- A bipartite graph is a graph with one group of m vertices and another group of n vertices such that each vertex is adjacent to all of the vertices in the other group and adjacent to none of the vertices in their own group. This graph is denoted by $K_{m,n}$.
- The utility graph is $K_{3,3}$.

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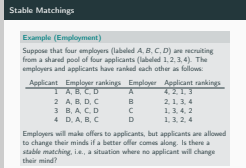
Example (Employment)

Suppose that four employers (labeled A, B, C, D) are recruiting from a shared pool of four applicants (labeled $1, 2, 3, 4$). The employers and applicants have ranked each other as follows:

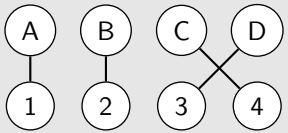
Applicant	Employer rankings	Employer	Applicant rankings
1	A, B, C, D	A	4, 2, 1, 3
2	A, B, D, C	B	2, 1, 3, 4
3	B, A, C, D	C	1, 3, 4, 2
4	D, A, B, C	D	1, 3, 2, 4

Employers will make offers to applicants, but applicants are allowed to change their minds if a better offer comes along. Is there a *stable matching*, i.e., a situation where no applicant will change their mind?



An example of an unstable matching is the following:

Figure 3: A matching that is not stable.



This is because A prefers 2 to 1 and 2 prefers A to B. Therefore, both are inclined to switch.

Stable Matching Algorithm

- The previous problem can be solved by an algorithm. For each round:
  1. Each employer with an open position makes an offer to the best candidate that they haven't made an offer to yet.
  2. Each applicant evaluates their offers and tentatively selects the best offer while rejecting the others.
  3. If there are any open positions remaining, then repeat this process in a new round of offers.

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This is known as the Gale-Shapley algorithm. It had previously been used to match residents with hospitals.